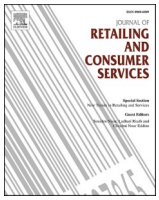
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Effects of crisis-induced inflation on purchasing and consumer behavior in Germany

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ABSTRACT

This study examines the impact of crisis-induced inflation and product scarcity on consumer behavior in Germany. Using the React-Cope-Adapt (RCA) framework, it analyzes how financial constraints and product shortages influence stress, financial planning, and purchasing decisions. The study is based on a two-wave survey (T1: N = 152, T2: N = 129) conducted during periods of high inflation and pre-Christmas economic conditions. Partial Least Squares Structural Equation Modeling (PLS-SEM) is applied to assess the dynamic interplay between financial constraints, stress, and behavioral adaptations over time.

The results indicate that financial constraints and product scarcity significantly increase stress levels, which in turn drive both impulsive (e.g., panic buying) and strategic (e.g., financial planning) consumer responses. Financial planning proves to be a key coping mechanism, leading to reduced discretionary spending and a greater reliance on private-label products. Over time, stress-driven behaviors, such as panic buying, decline, while structured financial adjustments persist.

The findings provide practical implications for policymakers and retailers. Transparent financial communication and targeted support measures can help mitigate stress, while retailers can address shifting consumer preferences by expanding cost-effective product offerings and strengthening supply chain resilience.

1. Introduction

Since 2020, German consumers have faced unprecedented challenges stemming from a series of global crises, which have collectively reshaped consumption and purchasing behavior. Among the key factors influencing these changes are inflation and international supply chain disruptions, both of which have been exacerbated by the COVID-19 pandemic and the Russia-Ukraine conflict. These crises have not only increased the cost of living but have also created significant scarcities in essential goods. Together, these developments have placed German households under substantial economic strain, forcing them to navigate a complex environment of financial constraints and product shortages simultaneously (Coleman and Nautz, 2023; Prohorovs, 2022; Tafa et al., 2022). Germany presents a particularly interesting case study due to its historically low inflation rates, its position as the largest economy in Europe and its critical role within the Eurozone, making the recent surge in inflation and its effects on consumer behavior and economic stability especially noteworthy (Glawe and Wagner, 2024).

Inflation, a central consequence of these crises, surged in Germany to levels not seen in decades. Fig. 1 illustrates the average annual inflation rates in Germany since the year 2000. Inflation in Germany remained relatively stable for decades, typically fluctuating around the European Central Bank's 2 % target. However, starting in 2021, inflation surged to levels exceeding 10 % in October 2022, marking the highest increase in over four decades (Federal Statistical Office of Germany, 2022). While inflation has moderated since August 2023, standing at 3.8 % in October 2023 and 2.3 % in January 2025 (Federal Statistical Office of Germany, 2025), it remains elevated compared to historical norms.

This inflationary period, driven by rising energy costs, supply chain disruptions, and geopolitical instability, significantly eroded purchasing power, with real wages dropping by over 4 %, the sharpest decline in nearly half a century (Federal Statistical Office of Germany, 2022). Research has shown that inflation expectations and credibility of monetary policy are central in shaping household financial behavior and economic stability (Coleman and Nautz, 2023). Inflation signifies a rise in the price level of goods and services over time, reducing the real value of material resources and highlighting monetary resource scarcity (Pitesa and Thau, 2014). High inflation rates undermine predictability

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and financial security, resulting in environmental uncertainty characterized by worry, fear, and a loss of control (Hamilton et al., 2019a). This uncertainty disproportionately affects lower-income households, leading to increased inequality in consumption patterns (Glawe and Wagner, 2024). Such conditions frequently alter consumer behaviors, including reduced expenditure on non-essential items and increased demand for cost-effective alternatives, such as private labels (Kaswengi and Diallo, 2015).

Compounding these financial pressures are the effects of supply chain disruptions. The COVID-19 pandemic caused widespread interruptions in global production and logistics, which were further aggravated by the geopolitical fallout from the Russia-Ukraine conflict. These disruptions led to production halts, delivery delays, and acute shortages of certain goods, particularly in critical sectors such as energy and food (Prohorovs, 2022). Product scarcity as a consequence of supply chain disruptions arises when systemic inefficiencies in production, transportation, and distribution impede the availability of goods (Orlando et al., 2022). Unlike temporary shortages caused by localized issues, the current scarcity reflects prolonged and widespread disruptions across global supply networks. Sector-specific studies highlight that supply chain resilience and alternative manufacturing approaches, such as additive manufacturing, could mitigate some of these challenges (Priyadarshini et al., 2023). For instance, the COVID-19 pandemic exposed vulnerabilities in just-in-time manufacturing models, leading to significant delays and backlogs in sectors such as electronics, automotive, and healthcare (Moosavi et al., 2022).

These challenges were further compounded by geopolitical instability, such as the Russia-Ukraine conflict, which disrupted critical supply lines for energy, grains, and raw materials essential to manufacturing and food production (Prohorovs, 2022). As a result, German consumers experienced not only higher prices but also reduced access to essential and discretionary goods (Coleman and Nautz, 2023; Tafa et al., 2022). These dynamics foster an environment where scarcity itself becomes a driver of inflation, creating a self-reinforcing loop that intensifies economic pressures on households. Moreover, scarcity influences consumer behavior beyond financial considerations, amplifying psychological factors such as urgency, perceived value, and decision-making under stress (Priyadarshini et al., 2023; Sheth, 2020). This dual burden forces households to make difficult trade-offs, such as prioritizing the acquisition of scarce goods at premium prices while cutting back on non-essential expenditures.

This study contributes to the literature on consumer behavior under financial and resource constraints by examining how crisis-induced inflation and product scarcity shape purchasing decisions over time. Unlike previous research, which has primarily focused on static effects of scarcity or inflation expectations (e.g., Coleman and Nautz, 2023; Sheth, 2020), our study introduces a dynamic perspective, applying the React-Cope-Adapt (RCA) framework (Hamilton et al., 2019a) to conceptualize consumer responses as an evolving process shaped by financial stress and adaptation strategies. This allows us to move beyond a simple cause-effect analysis. We instead demonstrate how coping mechanisms develop and shift in response to ongoing financial constraints. Additionally, we expand existing knowledge on financial planning as a structured coping mechanism during economic uncertainty. While previous studies have for example examined how inflation influences consumer behavior in general (e.g., Coleman and Nautz, 2023), our research empirically identifies how constrained consumers adjust their spending priorities over time, aligning with findings on the increasing relevance of cost-effective alternatives (Kaswengi and Diallo, 2015). By integrating insights from consumer psychology and behavioral economics, we provide a more nuanced understanding of how financial stress influences adaptive decision-making. From a practical perspective, the findings highlight the importance of transparent financial communication by policymakers to mitigate uncertainty and strengthen consumer confidence. Retailers can proactively respond by expanding private label offerings, implementing targeted pricing strategies, and ensuring supply chain resilience to stabilize purchasing behavior in times of financial distress. By linking empirical consumer insights with actionable policy and business implications, this study contributes to both consumer research and economic policy discussions.

Against this backdrop, the central research question of this study emerges:

What are the impacts of crisis-induced factors such as inflation and supply chain disruptions on purchasing and consumption behavior in Germany?

This article systematically examines the impacts of crisis-induced factors on consumer behavior in Germany. It begins by presenting the theoretical framework, focusing on the dual challenges of financial constraints and product scarcity, alongside the React-Cope-Adapt framework. Based on this, hypotheses are developed to explore the relationships between crisis factors and consumer behavior. The methodological section details a quantitative survey conducted at two time points, enabling a longitudinal analysis of behavioral changes under ongoing crises. The analysis and hypothesis testing section evaluates the empirical data, with findings discussed in relation to existing literature. This discussion highlights key insights, aligns or contrasts with prior research, and offers interpretations of observed behaviors. Implications for research, consumers and policymakers are provided, followed by a reflection on study limitations and suggestions for future research.

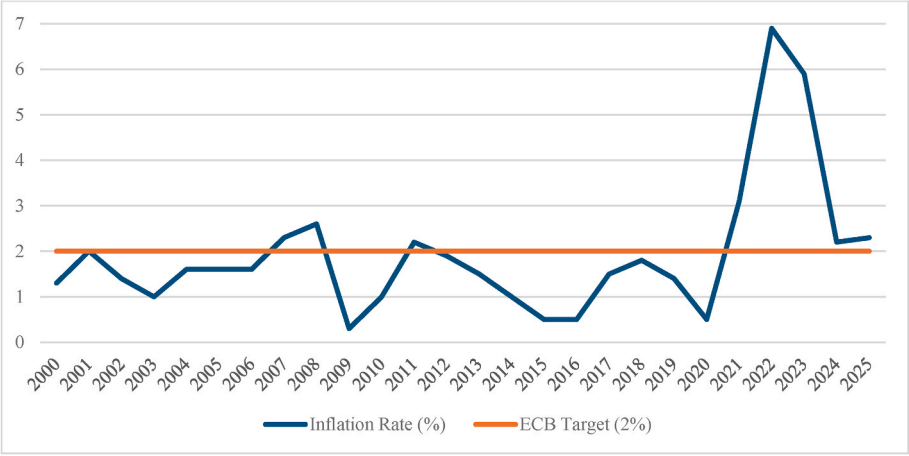


Fig. 1. Inflation trends in Germany 2000–2025 (Federal Statistical Office of Germany, 2025; Statista, 2025).

2. Theoretical foundation and hypotheses development

2.1. Theoretical foundation

Periods of historical inflation offer perspectives on the factors shaping current consumer behavior. For example, the 1973 oil crisis offers insights into how consumers adapted to inflationary pressures and scarcity, particularly in Germany, where a culture of financial caution has long influenced consumption decisions. The oil embargo imposed by OPEC countries during this period caused a significant surge in energy prices, leading to elevated production costs and a general rise in prices across the economy (Burdekin and Siklos, 1999). Understanding consumer responses to crises also requires a cultural lens, as reactions to scarcity and financial constraints are influenced by broader societal norms and values. Comparative studies, such as those examining consumer behavior across different economic contexts (e.g., Durante et al., 2015), have highlighted that cultural factors significantly mediate how consumers perceive and respond to resource constraints. In Germany, a strong tradition of financial prudence and risk aversion may influence consumer reactions differently than in economies with a more consumption-driven orientation. While our study does not engage in cross-cultural comparisons, this contextualization enhances the understanding of our findings and provides a foundation for future research exploring variations across different economic settings.

German consumers, shaped by historical experiences of hyperinflation in the 1920s and the economic crises following World War II, have a deeply ingrained propensity to avoid financial instability. During the inflationary period of the 1970s, households reduced their expenditure on durable goods and increased their savings (Hamilton, 1983). Furthermore, research has shown that expectations about future inflation and energy prices can amplify these behavioral shifts, influencing both consumption patterns and long-term financial planning (Filis et al., 2011).

Scarcity manifests in multiple forms. Financial scarcity arises when consumers lack sufficient resources to meet basic needs (Pitesa and Thau, 2014). A further significant dimension of scarcity is situational resource scarcity, which arises when life events (e.g., job changes, relocations) create discrepancies between available and required resources (Hosany and Hamilton, 2023). Scarcity also interacts with macroeconomic cues, such as recessions, influencing consumer behavior in nuanced ways (Durante et al., 2015). Moreover, Cannon et al. (2019) identify additional methodologies to study scarcity's impact, such as priming resource scarcity through exposure to scarcity-related words (Krosch and Amodio, 2014) or restricting product availability as part of scarcity marketing (Aggarwal et al., 2011). These methods reveal the cognitive and behavioral consequences of scarcity over time, illuminating how consumers navigate resource limitations. Both product and resource scarcity have been demonstrated to capture consumer attention, enhance the perceived value of items, and stimulate creativity in their use (Hamilton et al., 2019b). However, these forms of scarcity exert different effects. In the context of product scarcity, consumer choices are typically constrained. Conversely, resource scarcity may broaden consumer decision-making, prompting consideration of alternative options (Hamilton et al., 2019b). During crises, scarcity often drives extreme behaviors, such as panic buying and hoarding, as observed during the COVID-19 pandemic (Eger et al., 2021; Pantano et al., 2020).

Behavioral economics highlights how financial constraints and resource scarcity influence decision-making under uncertainty, often leading to systematic biases. A key concept in this context is mental accounting (Thaler, 1985), which explains how consumers categorize financial decisions into separate mental budgets. While this mechanism serves as a heuristic simplification in everyday life, it may lead to rigid spending patterns even under financial strain. In times of economic crisis, this behavior may contribute to misaligned resource allocation, where essential needs and discretionary expenses are not always balanced optimally.

Prospect Theory (Kahneman and Tversky, 2013) provides a further framework for understanding how individuals make decisions under uncertainty, particularly in contexts of financial scarcity. A central tenet of this theory is loss aversion, meaning that losses are psychologically more impactful than equivalent gains. This can lead consumers to avoid necessary spending adjustments or even take financial risks, such as accumulating debt, to maintain their accustomed standard of living. Additionally, availability bias can reinforce these behaviors, as consumers may rely on salient but potentially misleading information, such as media reports on inflation and scarcity, when making financial decisions. These mechanisms help explain why economic crises trigger behavioral patterns like panic buying or excessive saving, rather than rational budget adjustments.

The React-Cope-Adapt (RCA) framework, as proposed by Hamilton et al. (2019a), offers a comprehensive framework for analyzing consumer responses to financial and resource constraints as well as product scarcity. Hamilton et al. (2019a) originally developed the RCA framework in the context of financial constraints and resource scarcity, aiming to conceptualize how consumers dynamically adjust their behavior in response to uncertainty. The framework was first introduced to explain adaptive consumer strategies in situations where monetary or material resources are limited, emphasizing that coping mechanisms evolve over time rather than occurring as isolated responses. Since its introduction, the RCA framework has been applied in various contexts, including studies on consumption behavior during economic downturns, financial crises, and supply chain disruptions (Hamilton et al., 2019b; Sheth, 2020). Recent research has extended its application to crises such as the COVID-19 pandemic and inflationary pressures, demonstrating its relevance for understanding behavioral shifts under prolonged financial stress (Eger et al., 2021; Pantano et al., 2020).

The RCA framework was chosen in this study because it provides a structured, process-oriented approach to understanding consumer behavior during crises. Unlike static frameworks, it captures how consumers initially react, develop coping strategies, and ultimately adapt to financial and resource constraints, aligning well with our research focus. Thus, this framework conceptualizes consumer behavior as an evolving process through three sequential phases: React, Cope, and Adapt.

In the first phase (React), consumers primarily respond to perceived external stressors such as financial constraints and product scarcity. Financial constraints, characterized by limited resources, and the scarcity of certain goods due to supply chain disruptions or economic crises, heighten perceptions of uncertainty. During this phase, consumers focus on addressing these external challenges, often leading to impulsive, short-term-oriented decisions (Hamilton et al., 2019a; Sheth, 2020). As the crisis progresses, consumers enter the Cope phase, where emotional stress caused by perceived uncertainty plays a central role. To manage this stress and make better use of their limited resources, consumers begin to develop targeted financial planning strategies. These strategies include effective budget allocation, prioritization of essential needs, and careful planning of expenditures. Stress acts as a driver for structured and deliberate behavior, aimed at regaining control over the situation (Hamilton et al., 2019a; Ongkowijoyo et al., 2020). An important, yet often overlooked, emotional response during times of financial constraint is Financial Guilt. According to Martins et al. (2024), financial guilt can significantly shape consumer behavior, particularly when individuals perceive their consumption patterns as misaligned with broader economic responsibilities or personal financial goals. Dahl et al. (2003) further highlight that guilt in consumption contexts can manifest in different ways, including anticipated guilt before a purchase and post-purchase guilt, both of which may drive corrective behaviors. During crises, this form of guilt may exacerbate stress and influence both short-term reactions (e.g., panic buying) and long-term adaptations (e.g., increased savings or reduced discretionary spending). Consumers experiencing financial guilt might adjust their consumption behaviors not only out of necessity but also as a psychological response to alleviate feelings of regret or irresponsibility (Dahl et al., 2003). In the Adapt

phase, consumers establish new, sustainable consumption patterns based on their experiences with prior constraints and strategies. Reduced consumption becomes a long-term strategy to cope with financial limitations.

Similarly, the purchase of private labels gains prominence as a cost-effective alternative to premium brands. At the same time, stress-induced behaviors, such as panic buying and hoarding, persist, especially when uncertainty regarding the availability of specific resources remains high. These behaviors become more deeply embedded as part of an ongoing adaptation process (Sheth, 2020, Yurievna, 2022). Additionally, the concept of sustainability is critical in shaping these adaptive behaviors. Encouraging the consumption of durable goods can mitigate the impact of sudden economic crises, as consumers who invest in longer-lasting products may be less vulnerable to immediate financial shocks (e.g., Martins et al., 2024). This aligns with broader sustainability goals and reflects a shift towards more resilient consumption patterns.

The React-Cope-Adapt (RCA) framework thus illustrates how consumer behavior dynamically adjusts to external stressors. While the initial reaction is shaped by external constraints such as financial limitations and product scarcity, consumers progressively develop strategic approaches to deal with stress and limited resources. These strategies ultimately lead to a long-term transformation in consumption behavior, where frugality, the purchase of cost-effective alternatives, and persistent stress-driven behaviors, such as panic buying, become deeply integrated into daily life (Hamilton et al., 2019a; Sheth, 2020). In parallel, sustainability-oriented adaptations, such as prioritizing reusable products, adopting minimalistic consumption habits, and seeking more efficient resource use, can also emerge as coping mechanisms. These shifts suggest that financial constraints not only alter spending habits but can serve as a catalyst for a broader shift toward sustainable consumption behaviors. Consequently, this framework serves as the basis for subsequent hypothesis development, as it provides a foundation for understanding the dynamic interplay between financial constraints, stress, resource management, and behavioral adaptations.

2.2. Hypotheses development

In the React phase, consumers encounter external stressors such as financial constraints and product scarcity, which evoke emotional and impulsive responses. These stressors amplify perceptions of uncertainty, pushing consumers toward instinctive, survival-oriented behaviors (Hamilton et al., 2019a). Financial constraints, defined as a limitation in the availability of monetary resources, force individuals to make difficult trade-offs. For example, consumers must prioritize essential needs while foregoing discretionary items, often leading to feelings of anxiety and stress (Griskevicius et al., 2013; Shah et al., 2015). These decisions are not only resource-intensive but also emotionally taxing, as they highlight the challenge of maintaining stability in the face of scarcity. Similarly, product scarcity, which occurs when the availability of goods is limited, intensifies urgency and reallocates consumer focus toward securing scarce resources (Sharma and Alter, 2012). This scarcity is often heightened during crises, such as supply chain disruptions or economic downturns, creating a perception of competition among consumers. The COVID-19 pandemic provided a clear example: product scarcity, particularly for essential goods like toilet paper and food items, led to widespread panic buying and hoarding. These behaviors reflected a collective sense of losing control over the availability of critical resources (Eger et al., 2021; Lindenmeier et al., 2021).

Both financial constraints and product scarcity trigger emotional reactions, such as fear and stress, as consumers attempt to navigate these challenges. The unpredictability and perceived lack of control over external conditions heighten stress levels, which can further exacerbate impulsive behaviors and emotional decision-making. Thus, we propose the following hypotheses.

H1. Financial constraints lead to an increase in stress.

H2. Product scarcity leads to an increase in stress.

In the Cope phase, consumers shift from emotional and impulsive reactions to more deliberate strategies aimed at managing their limited resources and mitigating the psychological burden of stress. One of the most prominent strategies in this phase is financial planning, which becomes essential for adapting to resource constraints. Consumers engage in structured budgeting, prioritize spending on essential needs, and carefully plan their purchases to optimize the use of their available resources (Hamilton et al., 2019a). These behaviors reflect a calculated response to external pressures, as consumers strive to regain a sense of control over their financial situations. Research evidence supports this adaptive response. Constrained consumers are more likely to evaluate opportunity costs, consciously comparing trade-offs between competing expenses to maximize utility. This heightened focus on essential goods demonstrates a shift in decision-making priorities, where luxury or discretionary items are deprioritized in favor of necessities (Fernbach et al., 2015). For example, during economic downturns or crises, consumers have been observed to cut back on non-essential spending and actively seek cost-effective alternatives. Stress also plays a pivotal role in financial planning. High levels of stress, while often perceived as detrimental, can act as a trigger for deliberate and structured resource management. Under stress, consumers adopt coping mechanisms aimed at reducing uncertainty and restoring a sense of predictability in their financial decisions (Ongkowijoyo et al., 2020). For instance, stressed consumers are more likely to implement disciplined financial behaviors, such as meticulous budgeting, as a means of regaining control (Hamilton et al., 2019a). This highlights the dynamic interplay between emotional regulation and strategic behavior in the Cope phase, where stress not only influences but also catalyzes adaptive financial responses.

This phase demonstrates that both external constraints (financial limitations and product scarcity) and internal emotional factors (stress) drive increased financial planning efforts. These deliberate actions are central to consumers' ability to navigate the challenges posed by constrained environments. Thus, we propose the following hypotheses.

H3. Financial constraints lead to an increase in financial planning.

H4. Product scarcity leads to an increase in financial planning.

H5. Stress leads to an increase in financial planning.

In the Adapt phase, consumers internalize behavioral changes and develop sustainable patterns over time, driven by the prolonged influence of constraints and stress on their decision-making processes. These adjustments reflect the culmination of earlier coping strategies, resulting in more ingrained and deliberate consumption behaviors. A prominent behavioral shift in this phase is the adoption of reduced consumption as a long-term strategy to manage ongoing financial limitations. Consumers increasingly minimize discretionary spending, focusing on essentials to better align their expenses with constrained resources. This reduction in consumption is often accompanied by heightened sensitivity to value, as consumers carefully evaluate their spending choices and seek to eliminate unnecessary expenditures (Griskevicius et al., 2013). For example, during economic crises, reduced income or increased costs lead consumers to prioritize necessities while cutting back on luxuries or non-essentials.

Simultaneously, private labels gain significant prominence as cost-effective alternatives to premium brands. These products offer comparable utility at a lower cost, making them particularly appealing in financially constrained environments. Studies have consistently shown that during periods of economic uncertainty, consumers demonstrate a marked increase in their preference for private labels, reflecting a strategic response to financial constraints (Kaswengi, Fall and Diallo, 2015; Lamey et al., 2007). This shift underscores a deliberate adjustment in consumer behavior, where cost-saving measures become a key priority in maintaining financial stability. In addition to these rational behaviors, stress-induced responses such as panic purchases continue to persist

in this phase, particularly when uncertainty about the availability of resources remains high. Even as consumers adopt more structured financial strategies, the lingering effects of stress can trigger impulsive behaviors aimed at securing perceived scarce resources (Lindenmeier et al., 2021; Omar et al., 2021). For example, consumers may stockpile essential goods or make bulk purchases out of fear of future shortages, even when such actions may not align with long-term planning objectives.

This phase demonstrates the dual influence of rational adaptations, such as reduced consumption and increased reliance on private labels, alongside emotional behaviors like panic purchases, reflecting the complex interplay between strategic adjustments and stress-induced impulses. Thus, we propose the following hypotheses.

H6. Financial planning leads to

(a). reduced consumption and

(b). a higher preference for private labels.

H7. The higher the level of stress, the more consumers tend to perform panic purchases.

In the following Fig. 2, the overall model is visualized, providing a comprehensive view of the hypothesized relationships and illustrating how the React-Cope-Adapt framework can be used to conceptualize the dynamic interplay between stress, financial planning, and consumer behavior during times of financial constraints and product scarcity.

3. Methodology

3.1. Data collection

To obtain data for the overall study, participants were recruited at two different time points through various channels, including emails, social networks, topic-specific forums, personal referrals from friends and acquaintances, and local communities. To ensure heterogeneity, invitations were distributed across varied demographic groups, including different age groups, educational backgrounds, professional fields, and geographic locations, covering both urban and rural areas. While participants were not explicitly screened based on their responsibility for household financial decisions, the nature of the survey, focusing on purchasing behavior and financial planning, suggests that those actively engaged in these tasks were more likely to participate. The sample includes a diverse demographic range, ensuring a broad representation of consumer experiences. The data was collected twice, initially in May and June 2023 (*T1, N= 152*) during the period of high inflation and subsequently during the pre-Christmas period of 2023 (*T2, N= 129*), where the inflation was not as high as in May and June 2023. The two data collection waves involved different respondents rather than a repeated measurement of the same individuals. This repeated cross-sectional approach allows for the identification of broader consumer trends but does not track individual behavioral changes over time. The selection of these two time points was based on key economic and seasonal factors. The first wave (May–June 2023) was conducted during a period of high inflation, providing insights into consumer behavior under pronounced financial constraints. The second wave (November–December 2023) captured behavioral shifts as inflation moderated. This combination allows for an analysis of both crisis-driven and seasonal consumption dynamics. The original sample consisted of 304 responses for both surveys. Following the implementation of data cleaning procedures, including the removal of responses with excessively short completion times (under 4 min), indicating insufficient response diligence, the data was reduced to 281 responses for analysis. Since the questions related to the central constructs were implemented as mandatory, no missing values occurred in the analyzed variables. The participants' ages ranged from 17 to 75 years, with an average age of approximately 41 years. Of the participants, 175 identified as female, 104 identified as male, one identified as non-binary, and one did not specify their gender identity. The distribution of educational qualifications shows a broad spread across different levels, with a substantial proportion holding a general university entrance qualification (39.9 %) or a university degree (23.5 %) (see Table 1). Similarly, income is distributed across various categories, preventing extreme bias toward a specific income group. These factors indicate a sufficient level of representativeness for our sample within the scope of this study.

Table 1
Education and income distribution among respondents.

Education	N	Percent	Income	N	Percent
Lower secondary school diploma	13	4.6 %	>1000€	84	29.9 %
Secondary school diploma/ Intermediate school certificate	40	14.2 %	1000 - 1500€	29	10.3 %
University of Applied Sciences entrance qualification	47	16.7 %	1501 - 2000€	28	10.0 %
General university entrance qualification	112	39.9 %	2001 - 2500€	29	10.3 %
University degree/Higher education degree	66	23.5 %	2501 - 3000€	23	8.2 %
Doctorate/PhD	3	1.1 %	>3000€	72	26.6 %
			No information	16	5.7 %

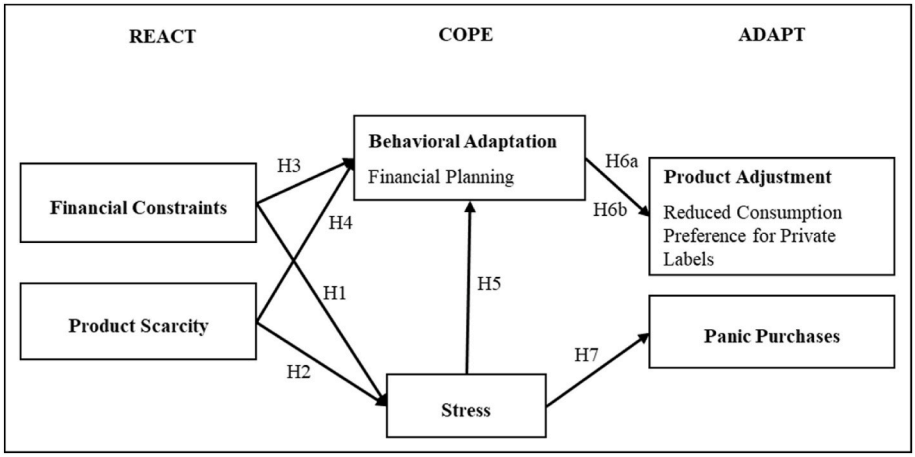


Fig. 2. Conceptual model.

3.2. Measurement model

The questionnaire for this study was designed to assess various aspects of purchasing and consumption behavior during the current crisis-induced inflation. The constructs were developed using validated measurement instruments originally sourced from English literature, which were then translated and partially adapted to fit the specific requirements of this research. The questionnaire employed a Likert scale format, offering seven response options ranging from 1 (Strongly Disagree) to 7 (Strongly Agree). To ensure questionnaire effectiveness, a pretest was conducted. The measurement of the constructs are displayed in Table 2.

Smart PLS4 was utilized to perform the analysis. The assessment of the measurement model involves evaluating construct reliability and validity. Construct reliability was determined using Cronbach's Alpha and Composite Reliability (*CR*). Table 3 presents the construct reliability and convergent validity for both the timepoint-specific sample and the full sample. Cronbach apla values and *CR* for all the constructs were higher than the recommendes value of 0.700 for the overall sample and the sample at the two different timepoints. The reliability and validity analysis shows predominantly good to excellent values for the scales, with the exception of the "Panic Purchases" scale at *T2*, which exhibits lower *CR* and *AVE* values.

As part of the structural model assessment, an important step is to assess the multicollinearity. We analyzed this using VIF. The VIF values met the recommended threshold of less than 5 (Hair et al., 2021) across the two time points and the overall sample, indicating no multi-collinearity issues, except for the construct "preference for private labels". Despite the elevated *VIF* values, the scale demonstrates high internal consistency (e.g., Cronbach's Alpha) and sufficient convergent validity ($AVE > 0.5$), indicating that the multicollinearity does not undermine the reliability and validity of the construct. Discriminant validity was assessed using the Fornell-Larcker criterion (Table 4) (Fornell and Larcker, 1981). According to this criterion, the square root of the AVE (diagonal, bold values) should be greater than the highest correlations between constructs (off-diagonal values).

The square roots of the AVE values are consistently higher than the highest correlations with other constructs. This confirms discriminant validity according to Fornell and Larcker (1981). The highest correlation (0.686 between Financial Planning and Reduced Consumption) remains below the corresponding AVE values (0.752 for Financial Planning and 0.764 for Reduced Consumption), indicating no critical discriminant validity concerns.

4. Results

4.1. Structural model

Before testing the hypotheses, we conducted a descriptive analysis of the model constructs to provide an overview of their central tendencies and variability. Table 5 presents the means (M) and standard deviations (SD) of the key constructs, measured on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). The results indicate that Financial Planning has the highest mean value, suggesting that participants tend to engage in financial planning behaviors more frequently. Similarly, Preference for Private Labels is relatively high, indicating a general inclination towards purchasing private label products. In contrast, Product Scarcity and Panic Purchases have lower mean values. Financial Constraints and Stress fall in the mid-range, implying a moderate level of financial strain and psychological distress within the sample.

A bootstrapping procedure with 5000 samples was performed to assess the significance of the path coefficients. In the initial analysis, we examined the direct relationships within the complete dataset. Fig. 3 indicates that all hypothesized relationships are significant. This confirms that our hypotheses can be supported for the entire sample. In the next step, a separate analysis of the relationships for the two data

Table 2
Source of measurement and items.

Construct	Item	Source
Financial Constraints	Compared to the time before high inflation … … my current financial situation is worse. … my current financial situation is better than that of my acquaintances. … I currently feel more financially constrained. … I currently feel less like I can spend as much money as I want. … my current financial situation is more restricted. … my household budget is more limited. … my household struggles to make ends meet over the month.	Ailawdi, Neslin, & Gedenk, 2001; Paley et al. (2019); Sharma and Alter (2012)
Reduced Consumption	Product durability is more important to me. I look for products that focus on essentials (e.g., squeegee/cloth instead of window vacuum). I purchase cheaper products than before. I buy fewer luxury items and more essentials.	Ang, 2000; Dangelico et al. (2022)
Preference for Private Labels	I buy more private-labeled products. I increasingly look for private-labeled items while shopping. My shopping cart usually contains several private-labeled products.	Ailawdi, Neslin, & Gedenk, 2001
Financial Planning	I increasingly compare prices of different brands before making a decision. Even for more affordable products, I currently check prices before purchasing. Getting products for the best possible price is more important to me right now. I'm currently being more mindful of my money. I'm currently buying more discounted items. I prioritize finding cheaper prices more than I used to.	Ailawdi, Neslin, & Gedenk, 2001 Ang, 2000; Shama (1981)
Product Scarcity	At the beginning of high inflation … … the products I wanted to buy were scarce. … particularly branded products were very scarce. … the package sizes I wanted to buy were scarce. … I couldn't afford the same products and brands as before the crisis.	Botti et al., (2008); Byun and Sternquist (2008); Yassin (2022); Hamilton et al. (2019a)
Stress	When I noticed that inflation had increased significantly, I felt worried … … about my well-being. … about my job. … about the economic situation. … due to a sense of loss of control.	Eger et al. (2021); Hamilton et al. (2019a); Larson and Shin (2018)

(*continued on next page*)

Table 2 (*continued*)

Construct	Item	Source
	… about the security of my future.	
Panic Purchases	At the beginning of high inflation …	Byun and Sternquist (2008); Yassin (2022)
	… I felt the need to buy certain products immediately.	
	… I wanted to purchase products during shopping that weren't planned.	
	… I spontaneously bought unplanned products.	
	… I bought more than I normally would have.	

collection time points was conducted, with the results presented in Table 6. At time point T1, the data suggest that all hypothesized relationships can be confirmed. However, at time point T2, a more nuanced picture emerges: while the majority of hypotheses remain supported at this time point, there are two exceptions. Hypothesis 4 (the effect of Product Scarcity on Financial Planning) and Hypothesis 7 (the effect of Stress on Panic Purchases) cannot be supported, as the corresponding relationships are not significant. This lack of significance suggests that the effects of these two variables have changed over time.

The Standardized Root Mean Square Residual (SRMR) for the estimated model was 0.085, slightly exceeding the threshold of 0.08 suggested by Hu and Bentler (1999), but still within a range often considered acceptable in exploratory research (Henseler et al., 2016). The slightly elevated SRMR value may be due to model complexity and temporal effects between T1 and T2. Since PLS-SEM is primarily prediction-oriented, such deviations are not uncommon. Overall, the model demonstrates sufficient explanatory power, as indicated by the R^2 and Q^2 values.

Based on the R^2 and Q^2 values (Table 6), the model explains the variance of most dependent variables moderately to well, particularly for Reduced Consumption and Stress, which show strong model fit and predictive power. In contrast, Panic Purchases exhibits low explanatory power and predictive accuracy, suggesting limited relevance of this variable in the model. All in all, this further evaluation of the fit measures and theoretical grounding supports the adequacy of the model, confirming its overall validity.

Furthermore, mediation analyses were conducted to examine whether Stress or Financial Planning function as mediators within the hypothesized relationships. The results indicate that both Stress and Financial Planning partially mediate the relationships between the independent and dependent variables, meaning that they contribute to explaining these associations but do not fully account for them. This suggests that both Stress and Financial Planning play a role in explaining the relationships between the independent and dependent variables. However, findings suggest that while Stress and Financial Planning provide additional explanatory value by partially mediating the relationships, they are not essential to establish the direct connections. Instead, the direct effects are strong enough to remain significant, even in the absence of the mediators. This underscores the robustness of the direct relationships, suggesting that the theoretical model primarily relies on direct influences rather than being fundamentally dependent on mediation effects. Thus, the mediation processes can be interpreted as complementary mechanisms that refine the understanding of these relationships rather than as necessary conditions for the effects to occur.

4.2. Multi-group analysis

In the final stage of the research, we analyzed if there were notable variations between time points *T1* and *T2* in the context of hypotheses testing. The findings indicate that the outcomes of most hypotheses remain consistent across the two time points, with the exception of H2 and H6a. Specifically, the difference in path coefficients shows that the impact of Product Scarcity on Stress (H2) is stronger at *T1* compared to *T2*. Additionally, the negative difference in the path coefficient for H6a

Table 5
Descriptive analysis of the constructs.

Construct	N	M	SD
Financial Constraints	281	4.35	1.57
Reduced Consumption	281	4.00	1.33
Preference for Private Labels	281	4.79	1.76
Financial Planning	281	5.14	1.25
Product Scarcity	281	3.35	1.46
Stress	281	4.29	1.52
Panic Purchases	281	3.46	1.56

Table 3
Reliability and convergent validity.

	Complete			*T1*			*T2*		
Items	*Alpha*	*CR*	*AVE*	*Alpha*	*CR*	*AVE*	*Alpha*	*CR*	*AVE*
Financial Constraints	0.912	0.933	0.699	0.923	0.941	0.727	0.896	0.921	0.663
Product Scarcity	0.800	0.870	0.629	0.789	0.865	0.618	0.816	0.881	0.652
Stress	0.850	0.893	0.625	0.843	0.888	0.615	0.858	0.897	0.636
Financial Planning	0.846	0.886	0.566	0.870	0.902	0.607	0.815	0.863	0.515
Reduced Consumption	0.768	0.846	0.583	0.763	0.841	0.574	0.775	0.853	0.595
Preference for Private Labels	0.945	0.965	0.901	0.928	0.955	0.876	0.963	0.976	0.932
Panic Purchases	0.830	0.885	0.659	0.863	0.907	0.710	0.776	0.609	0.373

Table 4
Fornell-Larcker matrix.

Construct	Financial Constraints	Financial Planning	Panic Purchases	Preference for Private Labels	Product Scarcity	Reduced Consumption	Stress
Financial Constraints	**0.836**	0.494	0.340	0.310	0.485	0.480	0.484
Financial Planning	0.494	**0.752**	0.249	0.570	0.384	0.686	0.448
Panic Purchases	0.340	0.249	**0.812**	0.108	0.379	0.257	0.378
Preference for Private Labels	0.310	0.570	0.108	**0.949**	0.272	0.552	0.242
Product Scarcity	0.485	0.384	0.379	0.272	**0.793**	0.426	0.467
Reduced Consumption	0.480	0.686	0.257	0.552	0.426	**0.764**	0.435
Stress	0.484	0.448	0.378	0.242	0.467	0.435	**0.791**

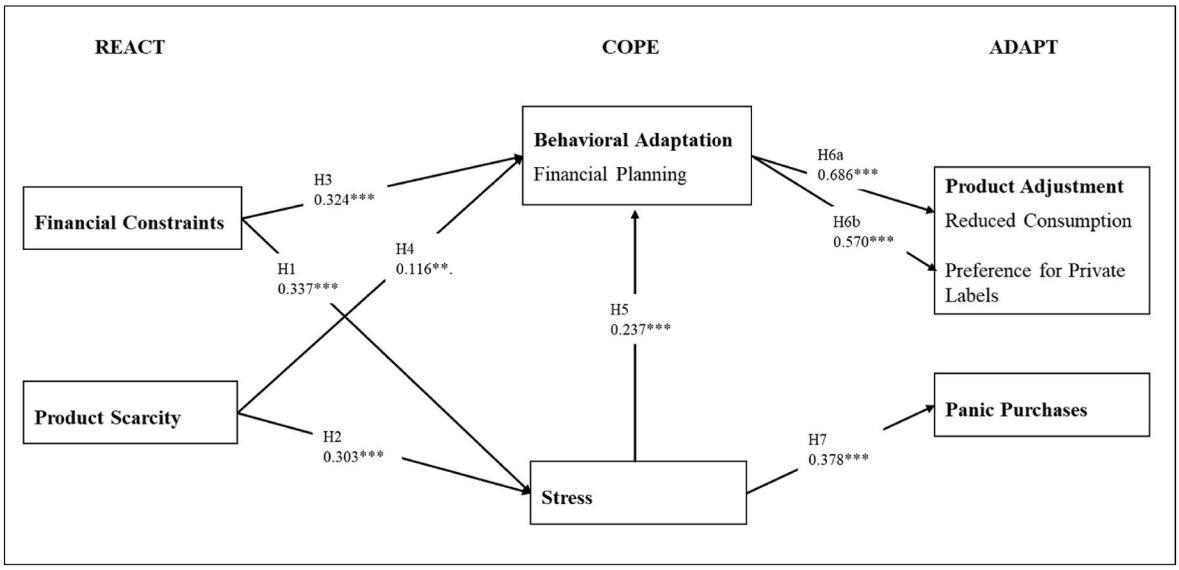


Fig. 3. Direct Relationships of the complete model.

Table 6
Direct relationships.

	Complete		*T1*		*T2*	
Hypotheses	*β*	*T*	*β*	*T*	*β*	*T*
H1: Financial Constraints - > Stress	0.337***	5.302	0.325***	4.092	0.267***	2.491
H2: Product Scarcity - > Stress	0.303***	5.344	0.408***	5.837	0.255***	2.782
H3: Financial Constraints - > Financial Planning	0.324***	4.685	0.34***	4.103	0.309***	2.686
H4: Product Scarcity - > Financial Planning	0.116**	2.167	0.144**	1.873	0.069 *n.s.*	0.755
H5: Stress - > Financial Planning	0.237***	3.289	0.228***	2.572	0.23**	1.941
H6a: Financial Planning - > Reduced Consumption	0.686***	20.412	0.660***	12.789	0.744***	27.821
H6b: Financial Planning - > Preference for Private Labels	0.570***	12.935	0.561***	9.064	0.599***	9.838
H7: Stress - > Panic Purchases	0.378***	7.272	0.536***	9.259	0.253 *n.s.*	0.939
	R^2	Q^2	R^2	Q^2	R^2	Q^2
Financial Planning	0.310***	0.259	0.364***	0.309	0.239***	0.158
Stress	0.305***	0.289	0.415***	0.393	0.199***	0.159
Reduced Consumption	0.470***	0.239	0.436***	0.292	0.553***	0.158
Preference for Private Labels	0.325***	0.108	0.315***	0.151	0.359***	0.047
Panic Purchases	0.143***	0.126	0.288***	0.270	0.064 *n.s.*	0.035

Note: *** if $p < 0.01$, ** if $p < 0.05$, * if $p < 0.10$., *n.s.* = not significant.

indicates that the influence of Financial Planning on Reduced Consumption is more pronounced at *T2* than at *T1*. These variations highlight temporal shifts in the strength of these specific relationships. The results of the multi-group analysis, which provide detailed insights into these differences, are summarized in Table 7.

In addition to hypothesis testing, further analyses were conducted to compare the constructs of interest between the two time points. An independent samples *t*-test was performed to examine differences in perceived product scarcity, stress, and panic buying between May/June 2023 and the pre-Christmas period. Product Scarcity was perceived as significantly more intense during the months of May and June 2023 compared to the pre-Christmas period ($M_{t1} = 3.58$ ($SD_{t1} = 1.47$), $M_{t2} = 3.08$ ($SD_{t2} = 1.40$), $t(279) = 2.89$, $p = 0.004$). Stress exhibited an upward trend over time, with participants reporting significantly higher levels of concern during the pre-Christmas period ($M_{t2} = 4.48$, $SD_{t2} = 1.43$) than in May and June 2023 ($M_{t1} = 4.12$ ($SD_{t1} = 1.58$), $t(279) = -2.01$, $p = 0.045$). Panic Buying also increased over time, as participants reported a greater tendency to purchase or hoard products in anticipation of the upcoming Christmas holidays ($M_{t2} = 3.68$, $SD_{t2} = 1.43$) compared to May and June 2023 ($M_{t1} = 3.27$ ($SD_{t1} = 1.65$), $t(279) = -2.20$, $p = 0.028$).

Table 7
Multi-group analysis.

Relationships	Difference (*T1* - *T2*)	*p*
H1: Financial Constraints - > Stress	0.058 *n.s.*	0.336
H2: Product Scarcity - > Stress	0.153*	0.090
H3: Financial Constraints - > Financial Planning	0.030 *n.s.*	0.423
H4: Product Scarcity - > Financial Planning	0.076 *n.s.*	0.261
H5: Stress - > Financial Planning	−0.001 *n.s.*	0.497
H6a: Financial Planning - > Reduced Consumption	−0.083*	0.069
H6b: Financial Planning - > Preference for Private Labels	−0.038 *n.s.*	0.327
H7: Stress - > Panic Purchases	0.283 *n.s.*	0.071

Note: *** if $p < 0.01$, ** if $p < 0.05$, * if $p < 0.10$., *n.s.* = not significant.

5. Discussion and conclusion

The results reveal that both financial constraints and product scarcity significantly influence consumer stress levels, financial planning, and purchasing behaviors, aligning with prior research on the behavioral impacts of crises (Hamilton et al., 2019a; Sheth, 2020). The RCA framework proved to be a useful framework for conceptualizing these dynamic responses, as evidenced by the significant support for most

hypothesized relationships. Both financial constraints and product scarcity were shown to increase stress levels, particularly during the initial phase of the crisis (*T1*). This finding aligns with research indicating that scarcity heightens perceptions of uncertainty and urgency, leading to elevated stress (Griskevicius et al., 2013; Shah et al., 2015). However, the reduced impact of product scarcity on stress at *T2* suggests that consumers may become desensitized to scarcity over time or that improvements in product availability moderated its psychological effects (Hamilton et al., 2019a). The study corroborates the assertion by Hamilton et al. (2019a) that stress can act as a mediator between external constraints (e.g., financial or product scarcity) and consumer adaptation strategies, however, the results showed that the mediation effect is not decisive, as the direct effects of e.g. stress on financial planning and panic purchases are significant. Financial planning emerged as a pivotal coping mechanism, significantly influenced by financial constraints, product scarcity, and stress. This finding aligns with the RCA framework's Cope phase, where consumers adopt structured approaches to regain control over their resources (Hamilton et al., 2019a). Previous studies have highlighted that during economic downturns, financial planning enables consumers to optimize their limited resources and prioritize essential needs (Fernbach et al., 2015; Ongkowijoyo et al., 2020). The data confirm this trend, with financial planning leading to reduced consumption and a greater reliance on private labels as cost-saving strategies.

The study further highlights the long-term adaptations consumers develop, such as reduced consumption and an increased preference for private labels. Although these behaviors were already evident at T1, the increased effect of financial planning on reduced consumption at T2 suggests that frugality has become more entrenched over time. Additionally, the sustained preference for private labels at T2 indicates that cost-conscious shopping behaviors persist beyond immediate economic constraints. These findings align with previous research highlighting the long-term nature of consumption adjustments following economic downturns (Kaswengi and Diallo, 2015; Lamey et al., 2007). However, the persistence of stress-driven behaviors, such as panic purchasing at *T1*, underscores the dual influence of rational adaptations and emotional responses during crises (Eger et al., 2021; Lindenmeier et al., 2021). Interestingly, panic purchasing was not significant at *T2*, suggesting a decline in stress-induced impulsivity as inflation moderated and product availability improved. The lower CR and AVE values for panic purchasing at T2 may reflect its situational nature, as external factors likely influenced response patterns over time. Future research should refine the construct's operationalization to enhance measurement consistency. The temporal analysis provided by the multi-group analysis reveals notable shifts in consumer behavior between *T1* and *T2*. For instance, the impact of product scarcity on financial planning diminished over time, indicating that consumers may adjust to scarcity through learned strategies or reduced urgency. Conversely, the increased influence of financial planning on reduced consumption at *T2* suggests that frugality becomes more entrenched as a long-term strategy, even as economic conditions stabilize. Moreover, it was found that perceived product scarcity was significantly more pronounced during the months of May and June 2023 than during the pre-Christmas period. This suggests that consumers, shortly after the sharp rise in inflation, were more affected by product scarcity and likely perceived it as an acute threat. In contrast, participants expressed significantly higher levels of stress during the pre-Christmas period compared to May and June 2023. Although inflation decreased after May/June 2023, the sharp spike in inflation seems to have persistently influenced consumers' environmental concerns and triggered an increased awareness of environmental issues. Moreover, there has been an increasing tendency towards panic buying over time, suggesting that consumers are more likely to purchase or hoard products in anticipation of the upcoming holidays. This may be due to the closure of supermarkets during the Christmas period.

Implications

A key finding of the study is that stress plays a decisive role in the adjustment of consumer behavior. Stress, triggered by financial constraints and product scarcity, influences both impulsive behaviors, such as panic buying, and strategic adaptations, such as financial planning. To reduce panic buying, policymakers should implement measures that alleviate citizens' feelings of loss of control. Campaigns providing transparent information on product availability and price trends can help reduce uncertainty. Additionally, targeted financial support measures, such as energy price caps or direct household subsidies, could mitigate financial constraints and lower stress levels (Hamilton et al., 2019a).

The increased consumer preference for private labels during the crisis underscores the need to offer affordable alternatives (Kaswengi and Diallo, 2015). Retailers can adjust their strategies by expanding their range of private label and low-cost products to meet the changing preferences of consumers. Furthermore, retailers should implement mechanisms to flexibly respond to demand spikes, such as panic buying. Measures such as purchase volume limits or the introduction of reservation systems can help ensure supply even during crises (Kirk and Rifkin, 2020). Expanding personalized discount offers and price comparison tools could also support consumers' need for financial planning and control. The study further reveals that financial planning is an effective coping strategy for addressing uncertainties and constraints. Consumers should be encouraged to optimize their spending through targeted budgeting, price comparisons, and the use of discounts. Raising awareness of the long-term benefits of private labels and DIY activities can also help compensate for feelings of loss of control and better manage financial constraints (Hamilton et al., 2019a). Additionally, promoting sustainable consumption, such as the deliberate avoidance of unnecessary products, could help reduce stress and conserve resources.

Beyond financial planning, product quality should be emphasized to reduce consumer reliance on panic-driven purchasing decisions. Retailers and policymakers should encourage the availability of durable goods, as high-quality products with longer lifespans reduce the need for frequent repurchasing and help consumers maintain stability in times of financial uncertainty. Ensuring that consumers recognize these benefits through clear labeling and consumer education campaigns could strengthen their adoption of quality-driven purchasing strategies. Moreover, product shelf life significantly influences purchase behavior during crises. Non-perishable goods, such as canned foods, dry staples or hygiene products, are particularly susceptible to stockpiling behaviors, as consumers seek to ensure availability over extended periods. Future policy measures could help mitigate excessive hoarding by promoting the consumption of longer-lasting alternatives and developing inventory strategies to maintain steady supply levels of essential goods. Retailers could also implement stock management systems to prevent shortages while providing clear product labeling to help consumers make informed purchasing decisions. Understanding the role of product longevity in financial planning could further enhance adaptive consumer behaviors and reduce unnecessary financial strain.

Limitations and future research directions

This study has certain limitations that future research should address. First, reliance on self-reported data may introduce biases, as participants might provide socially desirable responses or struggle to accurately recall their behaviors. Incorporating objective data sources, such as actual purchase records, could enhance the validity of future findings. Additionally, data collection was conducted at two specific points in time, corresponding to different phases of the crisis. While this approach allows for the observation of short- to medium-term behavioral adjustments, it does not constitute a fully longitudinal panel study. The repeated measurement design provides insights into behavioral adaptations during crisis periods but does not track dynamic long-term

transformations. Future research could extend this by incorporating multi-wave longitudinal designs to examine the persistence of behavioral changes beyond the immediate crisis context. Consequently, the question arises of how long the adaptive strategies developed during the crisis, such as reduced spending and an increased preference for private labels, persist once inflation stabilizes. Research by Lamey et al. (2007) suggests that such behaviors tend to decline gradually after an economic downturn. Longitudinal studies could further explore the longevity of these adjustments. Furthermore, the timing of the second data collection period (November–December 2023) presents a potential limitation. As this phase coincides with the pre-Christmas season, consumer behavior may additionally be influenced by seasonal shopping patterns and not solely reflect adjustments to economic constraints. This could particularly affect findings related to financial planning and spending priorities. While the second period was intentionally chosen for data collection to capture behavioral shifts as inflation pressures declined, it is possible that holiday-related factors introduced additional variability in consumption behavior. To better isolate the effects of financial constraints from seasonal influences, future research should consider incorporating additional time points beyond holiday seasons.

Moreover, the study primarily focuses on German consumers, which limits the generalizability of the findings. While this limitation is acknowledged, future research could strengthen the applicability of the results by conducting comparative studies across different cultural and economic contexts. Cross-cultural research has shown that consumer responses to financial constraints and scarcity can vary significantly depending on societal norms, institutional structures, and historical experiences with inflation (Durante et al., 2015). Expanding the analysis beyond Germany would help determine whether the observed behavioral patterns persist in other markets or if alternative coping strategies emerge. Thus, expanding the study to other markets would be valuable to account for potential cultural differences and enhance the external validity of the results. Although the sample size of $N = 281$ is appropriate for the analyses conducted and the multi-wave data collection strengthens methodological robustness, certain limitations remain with regard to our sample. Participants were recruited via various online channels, ensuring diversity in age, gender and income, yet potential self-selection bias cannot be fully ruled out. While online surveys are widely used in consumer research, regional and socio-economic variations may still affect generalizability. Future research could address this through randomized or quota-based sampling.

Another limitation is that the study primarily examines financial constraints in a broad sense. However, sudden external shocks, such as price fluctuations due to extreme weather events, strikes, or governmental policies, can lead to short-term panic buying and consumption shifts (Campbell et al., 2020). These unpredictable stressors often trigger stockpiling behaviors, causing an initial surge in purchases followed by a subsequent reduction, reflecting a prolonged stocking curve (Chandon and Wansink, 2002). While this phenomenon has been well-documented, particularly in the aftermath of crises like the COVID-19 pandemic (Campbell et al., 2020), its impact on consumer decision-making was not explicitly considered in this research. Future studies should examine how acute external events interact with long-term inflationary pressures to shape purchasing behavior over time. Additionally, research could explore how price competition mechanisms under stockpiling conditions influence consumer choices and market dynamics (Bell et al., 2002).

Furthermore, while this study focused on general financial constraints, future research should examine the effects of specific cost increases, such as rising gas or electricity prices, on consumer behavior. Additionally, the increasing digitalization of consumer habits profoundly impacts adaptation strategies. Exploring how technological tools, such as price comparison apps or online shopping platforms, facilitate consumer coping mechanisms during crises could offer valuable insights. Furthermore, financial technology solutions, such as AI-driven budget management applications, may help consumers mitigate financial stress by optimizing expenditure planning in real-time (Gomber et al., 2018). Integrating these technological aspects into future research could expand our understanding of how digital tools shape financial resilience.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used DeepL and ChatGPT to assist with translation from German to English, as we are non-native speakers. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Theresia Mennekes: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hanna Schramm-Klein:** Writing – review & editing, Supervision, Resources, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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